

The “Should I Sell?” Stock Dashboard Tool

Project Proposal

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Target Audience: Financial Advisors and their Clients

Technology Stack: Python, Power BI, Alpha Vantage API

Background

This project proposes the development of an interactive data dashboard, created in Power BI, that will utilize current and historical stock market data to provide insight stock price trends over time, to be used during times of market correction as a tool to help inform investment strategy. The dashboard will leverage the Alpha Vantage API to retrieve accurate historical stock market data, which will then be manipulated to create comparison points based on past price corrections to illustrate factors such as Percent Drop in Value, Time to Recovery, Value if Divested, Value Today if Not Divested. These data points can be used to assist financial advisors in illustrating their rationale for the investment strategy advice they offer.

Business Problem

Financial Advisors are not just managers of money, but also managers of their clients' emotions. Especially during times of economic turmoil, clients frequently approach their financial advisor looking to pull investments out of the stock market to limit losses; an understandable reaction that is largely driven by fear. This tool is meant to aid financial advisors in showcasing easy-to-understand evidence so that they may help address the emotionally driven inquiries of their clients. The hope is that by providing big-picture perspective on investment strategy through digestibly presented historical data trends, the financial advisor will have an easier time calming their clients' nerves and keeping them invested appropriately.

Objectives

- Build an interactive and insightful data dashboard in Power BI
- Utilize accurate current and historical stock data from Alpha Vantage API
- Develop data-driven recommendations on divest/invest/hold based on historical stock data
- Provide users with comparison points of price performance during previous periods of economic downturn
- Provide a foundation of data used to facilitate clients' emotional ease during economic stress.

Technical Approach

1. Data Integration and Processing

- **Stock Data Retrieval:** Connect to Alpha Vantage API to retrieve historical and real-time stock prices for chosen tickers.
- **Data Transformation:** Use Python scripts and/or power query in Power BI to process and structure data, calculating percent drops, recovery periods, and hypothetical divestment scenarios.
- **Scenario Modeling:** Simulate alternative outcomes; such as, selling at the bottom versus holding through recovery.

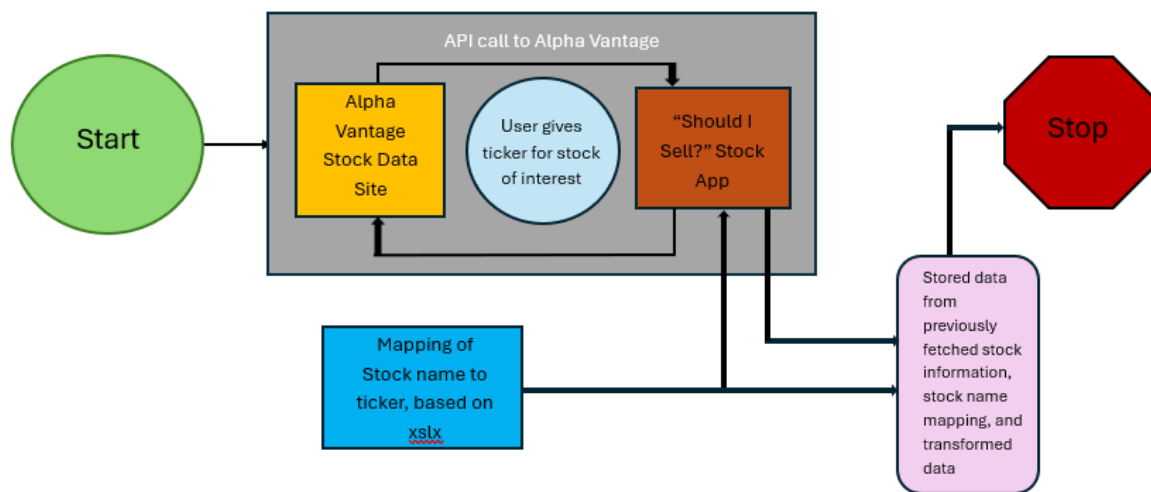
2. Recommendation Engine

- **Power BI Development:** Design intuitive charts and comparison tables showing:
 - Historical Corrections and recoveries
 - Value if divested at correction versus value if held through recovery
 - Key metrics such as percent drop, time to recovery, dividend impact
- **Activity Categories:**
 - Outdoor Sports (hiking, biking, boating)
 - Indoor Activities (museums, shopping, dining)
 - Weather-Dependent Events (picnics, festivals, outdoor concerts)
 - Seasonal Activities (fall foliage tours, winter sports)

3. Recommendation Engine

- **Historical Trend Analysis:** Provide present-day insights contextualized through historical data.
- **Evidence-Based Guidance:** Summarize insights for advisors to use in client conversations (ex: “Historically, holding through the correction resulted in XX% greater value after ___ many months.”)

Technical Architecture + Diagram



Backend Components

- **Data Retrieval:** Alpha Vantage API integration for stock data
- **Data Processing:** Pandas for data manipulation and transformation
- **Recommendation Engine:** Rule-based algorithm to recommend buy/hold/sell on stock of interest.
- **Data Storage:** Local CSV storage for stock data

Frontend Components

- **Web Framework:** Dash for web application development
- **Visualization:** Power BI for interactive charts and graphs
- **Styling:** A chosen financial advising firm can be chosen to source colors and fonts (i.e. Edward Jones, leads to yellows, greys, and blacks).

Data Flow

1. **API Integration:** Fetch and update stock data from Alpha Vantage API once daily.
2. **Data Processing:** Clean raw API data into user-friendly formats
3. **Data Calculation:** Transform Data into necessary Metrics through calculations.
4. **Recommendation Logic:** Suggest Buy/Sell/Hold based on algorithm logic.
5. **User Interface:** Display stock current/historical data and calculated hypotheticals through interactive dashboard
6. **User Interaction:** Allow filtering and drill downs based on user input.

Data Sources and APIs

Primary Data Source: Alpha Vantage API

- **Endpoint:** www.alphavantage.co/#about

Data Parameters:

- Stock ticker of users choice
- Time frame limited to present to last two recessions.
- United States Stock Market Only

Stock Name Mapping to Ticker Reference

- Excel-based lookup table mapping stock names and tickers
- Enables user to look up stock whether they know the name or the ticker

Activity Recommendation Logic

Stock Buy/Sell/Hold – Section with most True statements will result in that recommendation

1. Hold Indicators

- a) If hypothetical value today if held through recovery > selling at bottom of last recession
- b) If Dividends were maintained or increased during the last recession
- c) The company maintained similar or better leverage ratios during this point in time as the last recession.
- d) Operational cash flows remained resilient and stable during past recession periods
- e) Analyst ratings remained stable or improved following past recession recovery.

2. Buy Indicators

- a) IF stock price has already fallen the same percentage as the last recession AND
 - The stock price rebounded within one year of the last recession
 - The stock's sector historically outperforms the broader market during and after recession.
 - The company maintained similar or better leverage ratios during this point in time as the last recession.

3. Sell Indicators

- a) The stock underperformed or failed to recover after previous recessions.
- b) Dividends were reduced or suspended during the last downturn.
- c) The stock's sector lagged behind the broader market during and after recession periods.
- d) Debt-to-Equity ratio spiked, showing leverage increased dramatically, increasing risk of holding.

Responsive Design Considerations

- **Desktop:** Multiple charts displaying price trends over time, value of asset today, value if sold at bottom of last recession, value if held through recovery, and value if held through recovery until present. Value will be calculated based on input from user of number of shares owned.

Implementation Plan

Phase 1: Foundation & API Integration (Week 1)

- Establish project structure and define required stock market data fields
- Implement Alpha Vantage API integration for historical and current stock data retrieval
- Create initial stock data extraction scripts in Python
- Design the data processing pipeline for cleaning, transforming, and storing stock data
- Build stock ticker and company name lookup reference table

Phase 2: Core Features (Week 2)

- Develop key calculated metrics such as percent drop, time to recovery, value if sold, and value if held
- Build the initial rule-based buy/hold/sell recommendation engine
- Create core Power BI visuals for stock performance trends and recession comparisons
- Implement dashboard filters for ticker, recession period, and number of shares owned
- Add basic scenario modeling for divest-versus-hold outcomes

Phase 3: Enhancement & Refinement (Week 3)

- Refine recommendation logic using historical recession performance indicators
- Add supporting measures such as dividend impact, sector comparison, and leverage ratio analysis where available
- Improve dashboard usability and visual clarity for financial advisor presentations
- Optimize Python data transformations and Power BI model performance
- Begin user acceptance testing with sample advisor use cases

Phase 4: Testing & Optimization (Week 4)

- Conduct comprehensive testing of API retrieval, metric calculations, and recommendation outputs
- Validate dashboard calculations against known historical stock performance examples
- Fix bugs and improve performance in both the data pipeline and Power BI dashboard
- Gather user feedback on dashboard layout, clarity, and usefulness in advisor-client discussions
- Finalize supporting documentation for data sources, calculations, and logic

Phase 5: Deployment & Launch (Week 5)

- Publish the finalized dashboard to the production environment
- Establish a refresh schedule for stock data updates
- Monitor API reliability, data accuracy, and dashboard performance
- Create deployment and maintenance documentation
- Prepare recommendations for future enhancements, such as portfolio-level analysis or broader market benchmarking

Success Metrics

Technical Metrics

- **API Reliability:** >99% successful API calls
- **Page Load Time:** <3 seconds initial load
- **Data Accuracy:** Historical and current stock data matches source data from Alpha Vantage with no material discrepancies, up until time of data pull.

User Experience Metrics

- **User Engagement:** Average session duration >5 minutes
- **Feature Interaction:** More than 70% of users interact with filters, comparison visuals, or recommendation outputs
- **Recommendation Usefulness:** More than 70% of users report that the buy/hold/sell indicators support understanding of market downturn scenarios
- **User Satisfaction:** >4.5/5 star rating

Business Impact Metrics

- **Advisor Support Value:** Advisors report improved ability to explain investment strategy during periods of market volatility
- **Client Decision Support:** The dashboard helps facilitate more informed conversations around whether to buy, sell, or hold during downturns
- **Educational Value:** Clients gain a clearer understanding of how stocks have historically performed during and after recessions

Risk Assessment and Mitigation

Technical Risks

- **API Rate Limiting:** Alpha Vantage usage limits may restrict volume or frequency of stock data requests.
- **Data Accuracy:** Cross-reference other stock data sources if needed
- **Performance Issues:** Large datasets may slow down dashboard responsiveness. Mitigate this through preprocessing and limited scope of retrieved data to most relevant needs.

Project Risks

- **Scope Creep:** Expanding the project beyond its original object would delay completion.
- **Timeline Delays:** Data challenges, refinement, or testing issues could delay timelines, build in buffer periods into projected timeline.
- **Resource Constraints:** Certain desired measures, such as dividend history, sector benchmarking, or leverage ratios, may require additional sources beyond Alpha Vantage. To mitigate this, the initial version of the project will prioritize core stock price and recovery metrics, while documenting optional enhancements for future expansion.

Budget and Resources

Development Resources

- **Programming and Data Tools:** Dash, Pandas, Requests, PowerBI
- **Visualization Platform:** Power BI for dashboard development and interactive reporting
- **Data:** Alpha Vantage API (Free, Free API given at sign-up)
- **Design:** Dashboard styling aligned to a financial advisory use case, with a clean and professional layout

Time Investment

- **Development:** 5 weeks intensive development
- **Testing:** Integrated throughout each week
- **Documentation:** Ongoing throughout development
- **Maintenance:** Periodic updates for API monitoring, bug fixes, and future metric enhancements

Conclusion

The "Should I Sell?" Stock Dashboard Tool presents a unique opportunity to combine data science, financial analytics, and user-centric dashboard design to deliver a practical resource for financial advisors and their clients. By leveraging the Alpha Vantage API and open-source data processing libraries, the project demonstrates a cost-effective approach to developing impactful decision-support tools. The dashboard will provide advisors with evidence-based insights into historical stock performance during market downturns, supporting more informed and rational investment conversations. Ultimately, this solution aims to enhance advisor-client trust, reduce emotion-driven decisions, and promote better long-term outcomes.

Next Steps

1. **Stakeholder Review:** Present proposal to project sponsors and stakeholders
2. **Technical Feasibility Assessment:** Verify API capabilities and library compatibility
3. **Prototype Development:** Create minimum viable product (MVP) with core features
4. **User Research:** Gather feedback on activity preferences and UI requirements
5. **Project Kickoff:** Begin Phase 1 development with clear milestones and deliverables